#vlsi #characterization #FIB #ion_beam

focused ion beam

like a SEM but the beam is **ions** (usually gallium, Ga^{+}) instead of electrons. ions are ~100,000× heavier than electrons, so instead of just bouncing off they **knock atoms out** of the sample. it's a microscopic **knife**. part of Metrology and nano-characterization workshop

> **THE ONE IDEA**
>
> an electron beam looks, an ion beam cuts. same scanning idea as the SEM, but a heavy ion carries enough momentum to kick atoms out of the surface, so the FIB can image, mill, deposit and edit

why ions cut and electrons don't

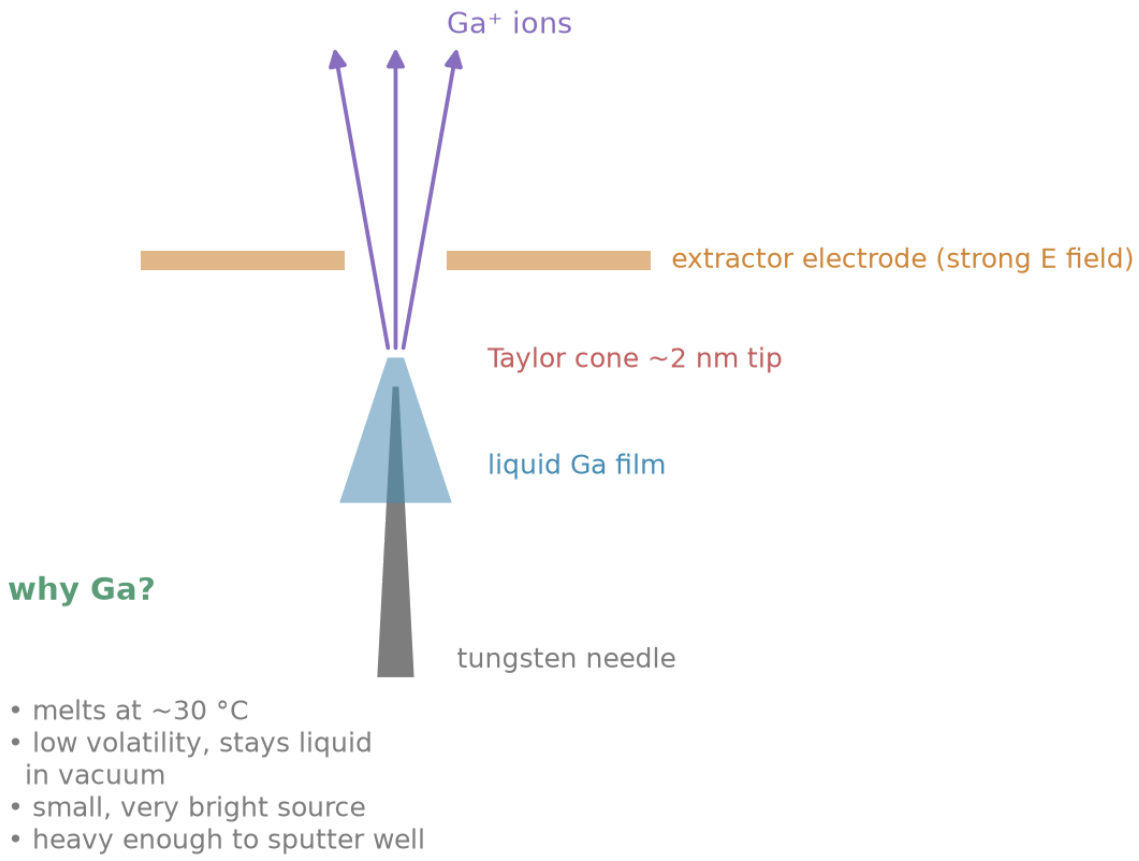
in a head-on collision, a moving mass m_1 with energy E can hand at most this much energy to a resting atom of mass m_2

$$E_{\max} = \frac{4m_1 m_2}{(m_1 + m_2)^2} E$$

it takes roughly **15–25 eV** to knock a Si atom out of its place (the displacement energy)

> **30 KEV GA ION VS 30 KEV ELECTRON HITTING SI ($m_{\mathrm{GA}} = 69.7$ U, $m_{\mathrm{SI}} = 28.1$ U, $m_e = 0.00055$ U)**
>
> $$\text{Ga:} \quad E_{\max} = \frac{4(69.7)(28.1)}{(69.7 + 28.1)^2} \times 30\ \mathrm{keV} = 0.82 \times 30\ \mathrm{keV} \approx 24.6\ \mathrm{keV}$$
> $$\text{electron:} \quad E_{\max} \approx \frac{4(0.00055)}{28.1} \times 30\ \mathrm{keV} \approx 2.3\ \mathrm{eV}$$
>
> the ion can hand over **a thousand times** the energy needed to knock an atom loose. the electron can't even reach the 15 eV, so an SEM looks without cutting (mostly. at a few hundred kV a TEM beam can start doing damage)

the source



liquid metal ion source (LMIS): a sharp tungsten needle wetted with liquid Ga. a strong electric field pulls the Ga into a tiny cone and ionizes it, giving a very small bright source (it's the ion version of the field emission gun)

- Ga because it's liquid near room temp (melts at ~30 °C), low volatility, low melting point
- lenses are **electrostatic** (ions are too heavy for magnetic ones to be convenient)

what the beam can do

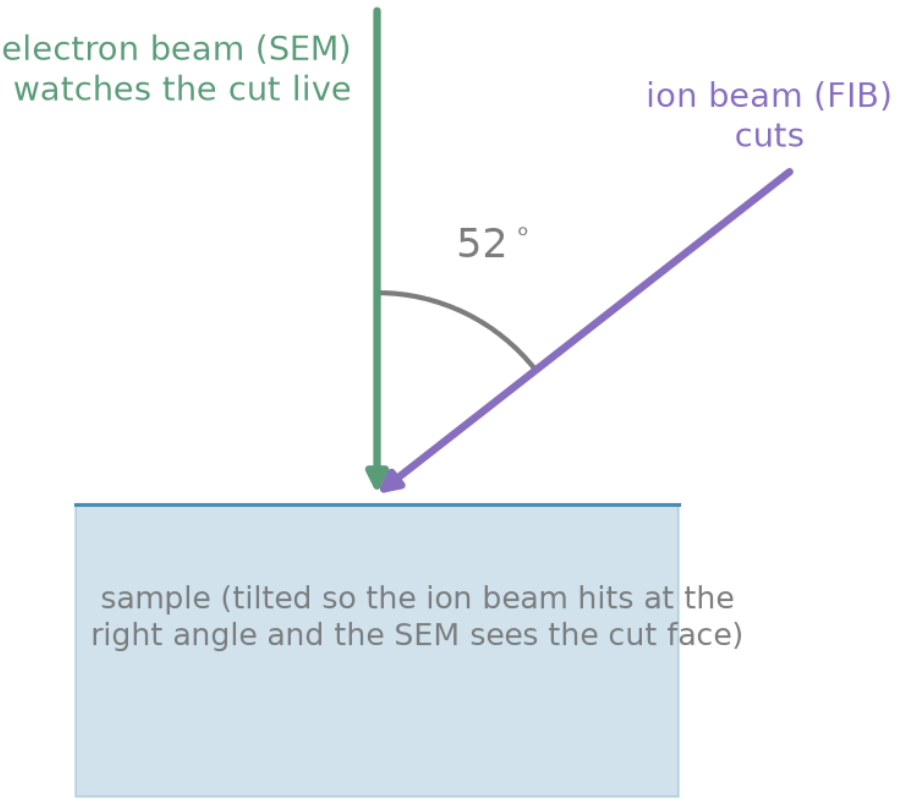
use	how
imaging	scan with a small current, collect ion induced secondary electrons
milling	high current, sputters atoms away → cut trenches, cross sections, holes
deposition	inject a precursor gas (eg. Pt, W, C) next to the beam, the beam cracks it and leaves a metal / insulator patch (**ion beam induced deposition**)
circuit edit	cut a metal line or add a wire to fix or probe a chip

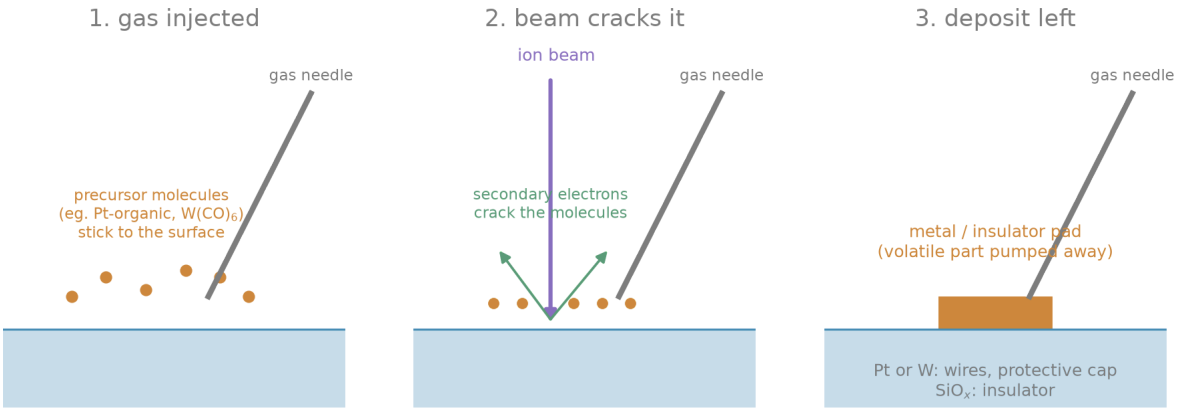
- beam current ranges from a few pA (imaging, fine polish) to tens of nA (fast bulk milling)
- imaging with ions always does a bit of damage, so you usually look with the electron beam (next section)

dual beam (FIB-SEM)

the standard machine has **both** columns pointing at the same spot (at an angle, usually 52°)

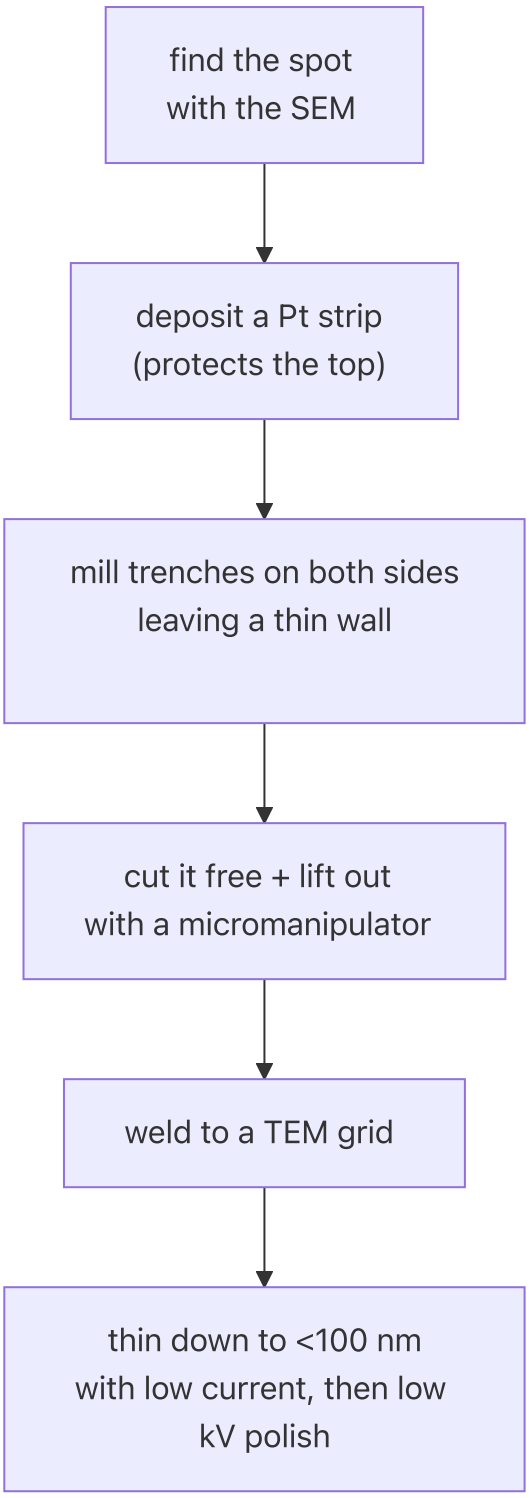
- ion beam cuts, electron beam **watches the cut live** without damaging the sample
- tilt the stage to look at the cross section face

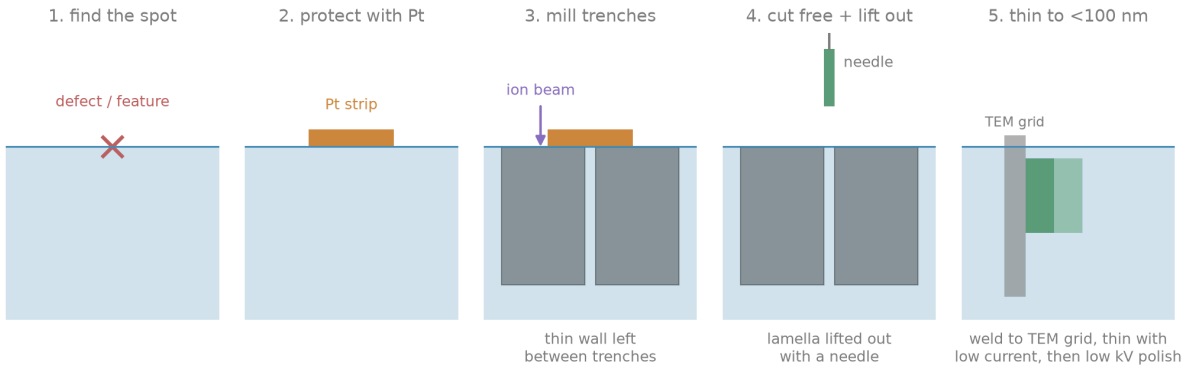




making a TEM lamella

the most common job: cutting a <100 nm slice from an exact spot so a TEM can look through it

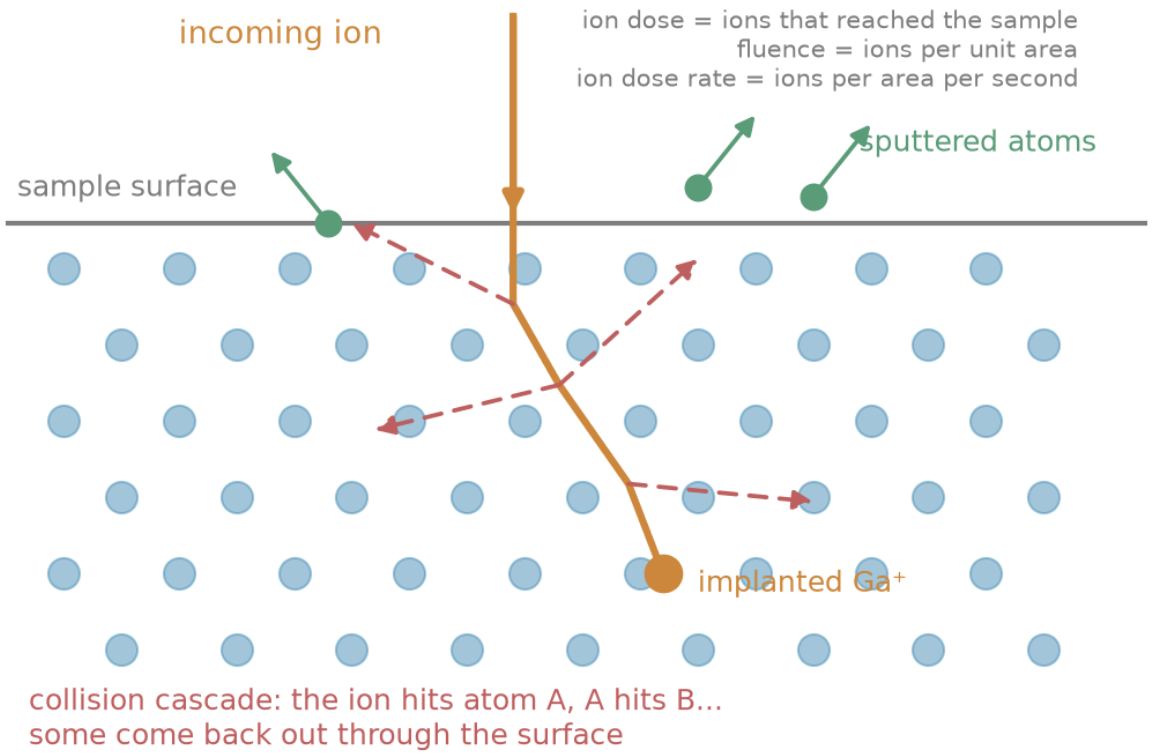

find the spot
with the SEM
deposit a Pt strip
(protects the top)
mill trenches on both sides
leaving a thin wall
cut it free + lift out
with a micromanipulator
weld to a TEM grid
thin down to <100 nm
with low current, then low
kV polish



what the ion does to the sample

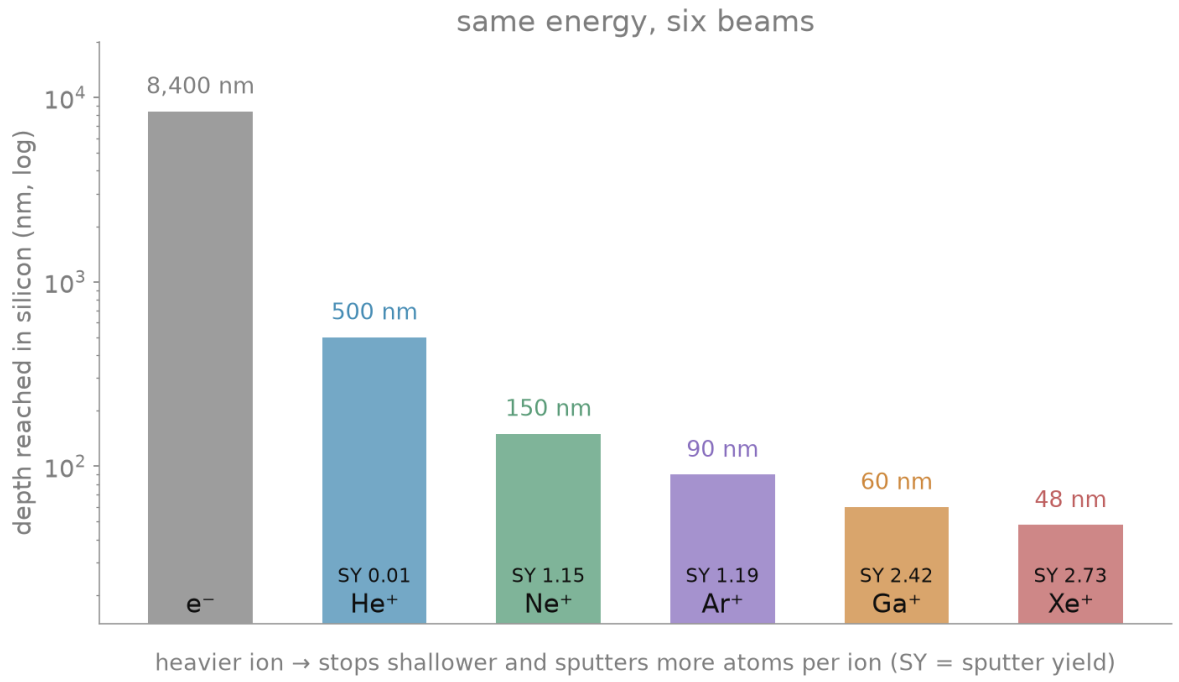
the ion hits, starts a **collision cascade**, atoms near the surface get knocked out (**sputtering**) and the ion ends up implanted

- **ion dose** = number of ions that reached the sample
- **fluence** = ions that pass through an area before the sample interaction
- **ion dose rate** = ions going through a specific area per unit time



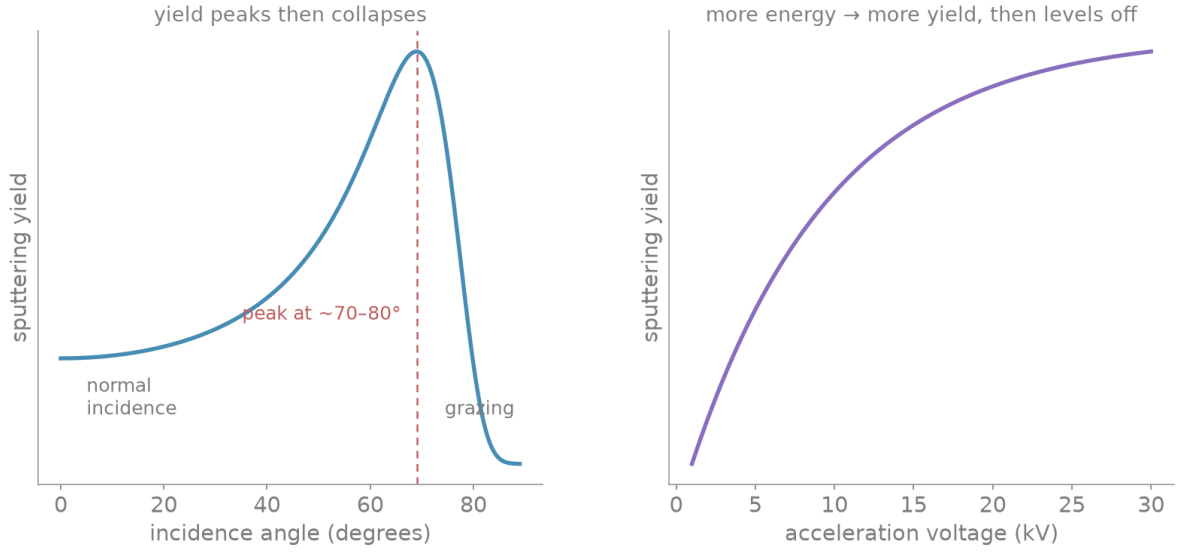
which ion?

at the same energy a heavier ion stops shallower and sputters more atoms per ion (sputter yield, SY)



parameters affecting sputtering

incidence angle, acceleration voltage and the surface binding energy of the material. yield peaks around 70–80° from the normal then collapses

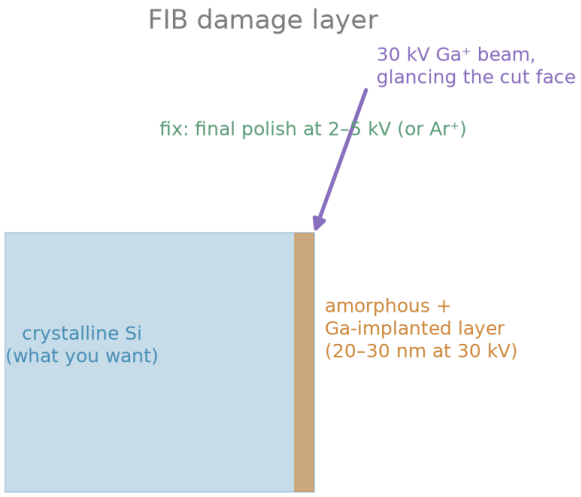


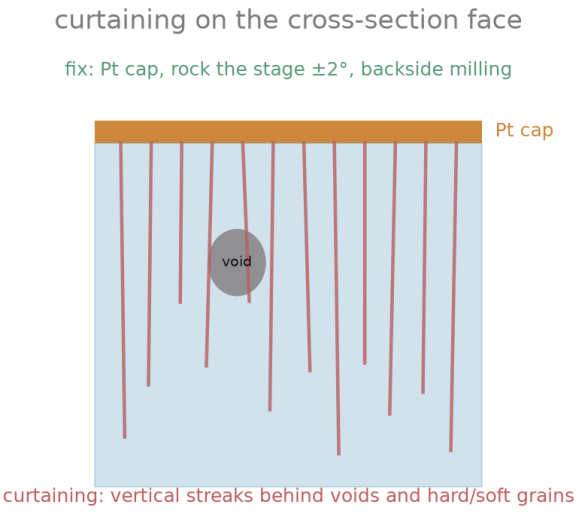
yield = atoms removed per incoming ion. Also depends on the surface binding energy of the material

downsides

- **Ga implantation** → Ga ends up in the surface
- **amorphous damage layer** → the top few nm of the surface get wrecked. fix with low kV final polish
- **redeposition** → sputtered material lands back nearby
- **curtaining** → vertical streaks from uneven material (protect with Pt, rock the sample)
- Ga can **react** with some materials (eg. Al, forms alloys)
- alternatives: **Xe plasma FIB** (much faster for big cross sections), He / Ne ion microscopes (imaging)

> ⚠ **THE DAMAGE LAYER SHOWS UP IN OTHER TECHNIQUES**
>
> a few nm of amorphous surface is invisible in an SEM, but it's a big deal for TEM (it blurs the lattice fringes) and it gives no peaks in XRD. that's why the last polish is done at low kV





artefacts and troubleshooting

Curtaining

cause: hard/soft phases, voids, topography

fix: smooth amorphous cap; rock sample; backside milling

Charging (SEM)

cause: insulators accumulate charge → drift, distortion, lost focus

fix: lower kV; deceleration; thin conductive coat

Redeposition

cause: sputtered material re-coats narrow trenches

fix: wider trenches; multipass scans; cleaning cross-sections

Drift during milling

cause: thermal drift, charging, stage creep → misplaced patterns

fix: equilibrate; shorter passes

Ga implantation & amorphization

cause: 20–30 nm damage at 30 kV (Si); Ga signal in EDS

fix: low-kV polish; Ar-ion polish

Channeling contrast

cause: grain brightness changes with orientation (real)

fix: interpret correctly; use BSE/EBSD to confirm

Beam-induced contamination

cause: hydrocarbons cracked by the beam build carbon deposits

fix: plasma-clean holder and sample; clean gloves; limit dwell

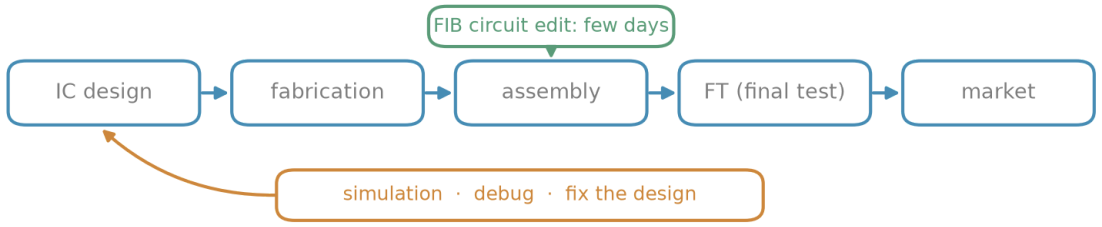
Rough walls / holes

cause: wrong overlap, too high final current or defocus

fix: correct focus at each current; always finish with a polish

case studies (from the lecture)

- **cross-section + TEM prep** of SiC MOSFETs (cracks between poly gate and Al source), dielectric breakdown spots, GaN HEMT failure (IDSS leakage from thermomechanical stress)
- **circuit edit** for rapid prototyping and failure correction (FIB nano-surgery, probe pads)



re-making the masks takes over a month (each rework)

FIB "nano-surgery": cut a metal line, add a wire (ion-beam-induced deposition), open a probe pad, then test the fixed chip right away, before touching the design masks

also: multi-point probe pads in specific areas for measurement

- **3D FIB-SEM tomography**: serial slicing and imaging, then reconstruct (eg. 3D NAND channel holes)
- **FIB-modified AFM probes** for trench profiles: sharpened tips for CD-AFM of fins and trenches, MEMS/NEMS sidewalls, 2D material steps

Helios 5 UX dual beam specs (CeNSE)

- electron column: Elstar, field emission, magnetic immersion lens, 0.6 nm at 30 kV (STEM), 0.7 nm at 1 kV, 0.8 pA–100 nA, 350 V–30 kV

- ion column: Phoenix, Ga, 1 pA–65 nA, 500 V–30 kV, 4.0 nm resolution at 30 kV
- detectors: in-lens SE/BSE (TLD, ICD, MD), Everhart-Thornley (ETD), in-chamber ICE (secondary ions and electrons), retractable STEM 3+ (BF/DF/HAADF)

in VLSI

- cross sections to look at layers inside a chip, **exactly where** a defect is
- TEM sample prep
- **failure analysis** and **circuit edit** → fix a design on silicon without remaking the masks
- 3D reconstruction (slice and view)

TRY THESE

① why is the ion column's resolution (4 nm) worse than the electron column's (0.7 nm), even though a Ga ion's wavelength is tiny?
② you need a lamella from one failed transistor among millions. why FIB and not an ultramicrotome?
③ use $E_{\max}$ to explain why a He ion microscope is gentler than a Ga FIB ($m_{\mathrm{He}} = 4.0$ u)

ANSWER

the quantum connection

MAKING QUBITS ATOM BY ATOM

a focused ion beam can also **implant** ions into a chosen spot, down to single ions. that's one way people place the atoms that act as qubits, eg. defect centres in diamond (a nitrogen next to a missing carbon) or single phosphorus donor atoms in silicon, where the qubit is the spin of one electron or nucleus. an alternative to the Superconducting qubits used by IBM and Google. the collision cascade above is exactly why it's hard: the ion stops at a **random** depth and leaves damage around it

FIXING AND INSPECTING QUANTUM CHIPS

FIB cuts the thin lamellae that let a TEM look inside a qubit's Josephson junction, and its circuit edit tricks (cut a line, deposit a new one) work on quantum chips the same way as on normal ones. the Ga it leaves behind matters more there though: stray atoms and damage are noise sources for qubits

see also SEM, TEM, XRD, ISS (the same collisions, used to measure), Metrology and

nano-characterization workshop

formulas: Microscopy formulas